

11th grade
Term 3 — Lesson 1 Practice Activity (C4) with Solutions

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Evaluated criteria

- C4: Applies the general procedure for hypothesis testing with fixed probability in contextual situations.

Problems

P1 1. Export compliance rate in a logistics hub

A logistics hub reports that at least 92 % of export containers pass customs compliance checks on the first inspection. A random sample of $n = 250$ containers shows $x = 219$ passes. At $\alpha = 0,05$, test whether the true pass proportion is lower than 0.92.

C4

- **Hypotheses:** $H_0: p = 0,92$, $H_1: p < 0,92$.
- **Conditions:** $np_0 = 250(0,92) = 230 \geq 10$, $n(1 - p_0) = 250(0,08) = 20 \geq 10$.
- **Sample proportion:** $\hat{p} = 219/250 = 0,876$.

■ **Test statistic:**

$$z = \frac{0,876 - 0,92}{\sqrt{\frac{0,92(0,08)}{250}}} \approx -2,57.$$

■ **p-value:** $P(Z \leq -2,57) \approx 0,0051$.

■ **Decision:** Reject H_0 because $0,0051 < 0,05$.

■ **Conclusion:** There is sufficient evidence that the first-pass compliance rate is below 92 %.

P2 2. Mobile banking adoption in rural districts

A policy team claims that the proportion of adults using mobile banking in rural districts is $p = 0,58$. From $n = 300$ surveyed adults, $x = 189$ report using mobile banking. At $\alpha = 0,10$, test whether adoption is greater than 0.58.

C4

■ **Hypotheses:** $H_0: p = 0,58$, $H_1: p > 0,58$.

■ **Conditions:** $np_0 = 174 \geq 10$, $n(1 - p_0) = 126 \geq 10$.

■ **Sample proportion:** $\hat{p} = 189/300 = 0,63$.

■ **Test statistic:**

$$z = \frac{0,63 - 0,58}{\sqrt{\frac{0,58(0,42)}{300}}} \approx 1,75.$$

■ **p-value:** $P(Z \geq 1,75) \approx 0,0401$.

■ **Decision:** Reject H_0 because $0,0401 < 0,10$.

■ **Conclusion:** There is sufficient evidence that rural mobile banking adoption is above 58 %.

P3 3. Defect proportion for exported microchips

A manufacturer states that the defect proportion of exported microchips is $p = 0,03$. In $n = 500$ chips, inspectors find $x = 25$ defects. At $\alpha = 0,01$, test whether the defect proportion differs from 0.03.

C4

■ **Hypotheses:** $H_0: p = 0,03$, $H_1: p \neq 0,03$.

■ **Conditions:** $np_0 = 15 \geq 10$, $n(1 - p_0) = 485 \geq 10$.

■ **Sample proportion:** $\hat{p} = 25/500 = 0,05$.

■ **Test statistic:**

$$z = \frac{0,05 - 0,03}{\sqrt{\frac{0,03(0,97)}{500}}} \approx 2,62.$$

■ **p-value:** $2P(Z \geq 2,62) \approx 0,0088$.

■ **Decision:** Reject H_0 because $0,0088 < 0,01$.

■ **Conclusion:** There is sufficient evidence that the defect proportion is not 3 %.

P4 4. Household electricity access program

A development report claims that 74% of households in a region have reliable electricity access, $p_0 = 0,74$. A sample of $n = 280$ households finds $x = 191$ with reliable access. At $\alpha = 0,05$, test whether the true proportion is below 0.74.

C4

- **Hypotheses:** $H_0: p = 0,74$, $H_1: p < 0,74$.
- **Conditions:** $np_0 = 207,2 \geq 10$, $n(1 - p_0) = 72,8 \geq 10$.
- **Sample proportion:** $\hat{p} = 191/280 \approx 0,682$.
- **Test statistic:**

$$z = \frac{0,682 - 0,74}{\sqrt{\frac{0,74(0,26)}{280}}} \approx -2,21.$$

- **p-value:** $P(Z \leq -2,21) \approx 0,0136$.
- **Decision:** Reject H_0 because $0,0136 < 0,05$.
- **Conclusion:** The sample supports that reliable electricity access is below 74 %.

P5 5. On-time repayment in microcredit portfolios

A lender expects an on-time repayment rate of $p_0 = 0,88$. In $n = 320$ loans, $x = 293$ are repaid on time. At $\alpha = 0,10$, test whether the repayment rate is different from 0.88.

C4

- **Hypotheses:** $H_0: p = 0,88$, $H_1: p \neq 0,88$.
- **Conditions:** $np_0 = 281,6 \geq 10$, $n(1 - p_0) = 38,4 \geq 10$.
- **Sample proportion:** $\hat{p} = 293/320 \approx 0,916$.
- **Test statistic:**

$$z = \frac{0,916 - 0,88}{\sqrt{\frac{0,88(0,12)}{320}}} \approx 1,99.$$

- **p-value:** $2P(Z \geq 1,99) \approx 0,0466$.
- **Decision:** Reject H_0 because $0,0466 < 0,10$.
- **Conclusion:** There is evidence that the repayment rate differs from 88 %.

P6 6. Inflation survey: households expecting higher prices

An economic institute states that 55% of households expect higher inflation next year. In a sample of $n = 360$, $x = 183$ households share that view. At $\alpha = 0,05$, test whether the true proportion is lower than 0.55.

C4

- **Hypotheses:** $H_0: p = 0,55$, $H_1: p < 0,55$.
- **Conditions:** $np_0 = 198 \geq 10$, $n(1 - p_0) = 162 \geq 10$.
- **Sample proportion:** $\hat{p} = 183/360 \approx 0,508$.
- **Test statistic:**

$$z = \frac{0,508 - 0,55}{\sqrt{\frac{0,55(0,45)}{360}}} \approx -1,61.$$

- **p-value:** $P(Z \leq -1,61) \approx 0,0537$.
- **Decision:** Fail to reject H_0 because $0,0537 > 0,05$.
- **Conclusion:** There is not enough evidence at 5 % to conclude the proportion is below 55 %.

P7 7. Digital tax filing completion rate

A ministry target is that 81 % of firms file taxes digitally, $p_0 = 0,81$. A sample of $n = 410$ firms finds $x = 349$ digital filings. At $\alpha = 0,01$, test whether the true proportion is greater than 0.81.

C4

- **Hypotheses:** $H_0: p = 0,81$, $H_1: p > 0,81$.
- **Conditions:** $np_0 = 332,1 \geq 10$, $n(1 - p_0) = 77,9 \geq 10$.
- **Sample proportion:** $\hat{p} = 349/410 \approx 0,851$.
- **Test statistic:**

$$z = \frac{0,851 - 0,81}{\sqrt{\frac{0,81(0,19)}{410}}} \approx 2,10.$$

- **p-value:** $P(Z \geq 2,10) \approx 0,0179$.
- **Decision:** Fail to reject H_0 because $0,0179 > 0,01$.
- **Conclusion:** Evidence is insufficient at 1 % to claim digital filing is above 81 %.

P8 8. Port cargo inspections with irregular documentation

Port authorities report an irregular-documentation proportion of 12 %, $p_0 = 0,12$. In $n = 450$ inspected shipments, $x = 69$ are irregular. At $\alpha = 0,05$, test whether this proportion has changed.

C4

- **Hypotheses:** $H_0: p = 0,12$, $H_1: p \neq 0,12$.
- **Conditions:** $np_0 = 54 \geq 10$, $n(1 - p_0) = 396 \geq 10$.
- **Sample proportion:** $\hat{p} = 69/450 \approx 0,153$.

■ **Test statistic:**

$$z = \frac{0,153 - 0,12}{\sqrt{\frac{0,12(0,88)}{450}}} \approx 2,18.$$

■ **p-value:** $2P(Z \geq 2,18) \approx 0,0292$.

■ **Decision:** Reject H_0 because $0,0292 < 0,05$.

■ **Conclusion:** The irregular-documentation proportion is significantly different from 12%.

P9 9. Graduate employment within six months

A university reports that 67% of graduates find jobs within six months, $p_0 = 0,67$. From $n = 260$ graduates, $x = 160$ are employed within six months. At $\alpha = 0,10$, test whether the true proportion is lower than 0.67.

C4

■ **Hypotheses:** $H_0: p = 0,67$, $H_1: p < 0,67$.

■ **Conditions:** $np_0 = 174,2 \geq 10$, $n(1 - p_0) = 85,8 \geq 10$.

■ **Sample proportion:** $\hat{p} = 160/260 \approx 0,615$.

■ **Test statistic:**

$$z = \frac{0,615 - 0,67}{\sqrt{\frac{0,67(0,33)}{260}}} \approx -1,88.$$

■ **p-value:** $P(Z \leq -1,88) \approx 0,0301$.

■ **Decision:** Reject H_0 because $0,0301 < 0,10$.

■ **Conclusion:** There is sufficient evidence that six-month employment is below 67%.

P10 10. Urban household recycling participation

A city plan states that 48% of households participate in weekly recycling, $p_0 = 0,48$. A sample of $n = 340$ households finds $x = 147$ participants. At $\alpha = 0,01$, test whether participation is lower than 0.48.

C4

■ **Hypotheses:** $H_0: p = 0,48$, $H_1: p < 0,48$.

■ **Conditions:** $np_0 = 163,2 \geq 10$, $n(1 - p_0) = 176,8 \geq 10$.

■ **Sample proportion:** $\hat{p} = 147/340 \approx 0,432$.

■ **Test statistic:**

$$z = \frac{0,432 - 0,48}{\sqrt{\frac{0,48(0,52)}{340}}} \approx -1,77.$$

■ **p-value:** $P(Z \leq -1,77) \approx 0,0384$.

■ **Decision:** Fail to reject H_0 because $0,0384 > 0,01$.

- **Conclusion:** At the 1 % level, evidence is insufficient to conclude recycling participation is below 48 %.